

General Definition of Expectations and an Application on Devil's Staircase

Special Mathematics Lectures – Introduction to probability

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1 Introduction

In the lectures [1], we have seen the definition of the expectation of discrete and absolutely continuous random variables. Recall that

$$\mathbb{E}(X) = \begin{cases} \sum_{x \in \text{Ran}(X)} x p_X(x) & \text{if } X \text{ is discrete,} \\ \int_{-\infty}^{\infty} x f_X(x) dx & \text{if } X \text{ is absolutely continuous.} \end{cases} \quad (1)$$

However, this definition relies on the probability mass function (pmf) p_X or the probability density function (pdf) f_X . For a general random variable X , we are only given the cumulative distribution function (cdf) F_X . In this case, how can we compute $\mathbb{E}(X)$, or more generally $\mathbb{E}(g(X))$? This is the question we will answer in this report.

2 Lebesgue-Stieltjes integral

Let us consider the measurable space $(\mathbb{R}, \mathcal{A}_B)$ where $\mathcal{A}_B$ is the *Borel σ -algebra* on $\mathbb{R}$. Recall that $\mathcal{A}_B$ is the smallest σ -algebra containing all intervals of $\mathbb{R}$. Note also that each element of $\mathcal{A}_B$ is called a *Borel set*. Next, consider a function $F : \mathbb{R} \rightarrow \mathbb{R}$ satisfying the following properties:

- (i) F is increasing: $x \leq y \Rightarrow F(x) \leq F(y)$,
- (ii) F is right continuous: $F(x) = F(x+0) := \lim_{\varepsilon \searrow 0} F(x+\varepsilon)$ for all $x \in \mathbb{R}$,
- (iii) $F(-\infty) := \lim_{x \rightarrow -\infty} F(x) = 0$ and $F(+\infty) := \lim_{x \rightarrow +\infty} F(x) < \infty$.

In the lectures [1], we define a *Stieltjes measure* m_F from the function F by starting with its values on half-open intervals:

$$m_F((a, b]) := F(b) - F(a), \quad \forall a, b \in \mathbb{R}, a < b. \quad (2)$$

Then, we extend this definition to all Borel sets in $\mathcal{A}_B$ by considering *coverings*. For any Borel set $V \in \mathcal{A}_B$, a covering of V is a family of half-open intervals $\{J_k\}$ so that $V \subset \bigcup_k J_k$. We then define

$$m_F(V) := \inf \left\{ \sum_k m_F(J_k) \mid \{J_k\} \text{ is a covering of } V \right\}. \quad (3)$$

Hence, we have defined a *measure space* $(\mathbb{R}, \mathcal{A}_B, m_F)$. For a Borel set $\Omega \in \mathcal{A}_B$, a function $g : \Omega \rightarrow \mathbb{R}$ is said to be *measurable* if $g^{-1}(V) \in \mathcal{A}_B$ for all Borel sets $V \in \mathcal{A}_B$. For a measurable

function g , we can define an integral of g with respect to the Stieltjes measure m_F . To do this, we start with a *simple function* $h : \Omega \rightarrow \mathbb{R}$ of the form

$$h = \sum_{k=1}^n a_k \mathbf{1}_{A_k}, \quad (4)$$

where $a_k \in \mathbb{R}$ and $A_k \in \mathcal{A}_B$ such that $A_k \cap A_j = \emptyset$ for all $k \neq j$ and $\bigcup_{k=1}^n A_k = \Omega$. Recall that $\mathbf{1}_{A_k}$ is the indicator function of A_k , i.e.

$$\mathbf{1}_{A_k}(x) = \begin{cases} 1 & \text{if } x \in A_k, \\ 0 & \text{if } x \notin A_k. \end{cases} \quad (5)$$

Since $h^{-1}(V)$ for any Borel set $V \in \mathcal{A}_B$ is a finite union of Borel sets A_k , the function h is measurable. The integral of h with respect to m_F is defined as

$$\int_{\Omega} h(x) dF(x) := \sum_{k=1}^n a_k m_F(A_k). \quad (6)$$

To explain the notation $dF(x)$, suppose that $A_k = (x, x+dx]$ is an interval with an infinitesimal length dx . Then, we have $m_F(A_k) = F(x+dx) - F(x) = dF(x)$. As a result, we have $a_k m_F(A_k) = h(x) dF(x)$ for $x \in A_k$.

For a non-negative measurable function $g : \Omega \rightarrow \mathbb{R}$, we can define its integral with respect to m_F by approximating g with simple functions. More precisely, we define

$$\int_{\Omega} g(x) dF(x) := \lim_{n \rightarrow \infty} \int_{\Omega} g_n(x) dF(x), \quad (7)$$

where $\{g_n\}$ is a sequence of simple functions such that $0 \leq g_n(x) \leq g(x)$ and $\lim_{n \rightarrow \infty} g_n(x) = g(x)$ for all $x \in \Omega$. For a general measurable function $g : \Omega \rightarrow \mathbb{R}$, we decompose g into its positive and negative parts: $g = g^+ - g^-$ where $g^+(x) = \max\{g(x), 0\}$ and $g^-(x) = \max\{-g(x), 0\}$. Then, we can define the integral of g as

$$\int_{\Omega} g(x) dF(x) := \int_{\Omega} g^+(x) dF(x) - \int_{\Omega} g^-(x) dF(x). \quad (8)$$

If we consider a complex-valued function $g : \Omega \rightarrow \mathbb{C}$ whose real and imaginary parts are measurable, we can define its integral by

$$\int_{\Omega} g(x) dF(x) := \int_{\Omega} \operatorname{Re}(g(x)) dF(x) + i \int_{\Omega} \operatorname{Im}(g(x)) dF(x). \quad (9)$$

The integrals we have defined so far are called the *Lebesgue-Stieltjes integrals* [2, Section 1.4].

Finally, if g is defined on the entire real line $\mathbb{R}$, i.e. $\Omega = \mathbb{R}$, then the expectation $\mathbb{E}(g(X))$ for a general random variable X can be defined as the Lebesgue-Stieltjes integral of g with respect to the cdf $F_X : \mathbb{R} \rightarrow \mathbb{R}$, i.e.

$$\mathbb{E}(g(X)) := \int_{\mathbb{R}} g(x) dF_X(x). \quad (10)$$

If X is a discrete random variable, then m_{F_X} is determined by its values on $\{x\}$ for $x \in \operatorname{Ran}(X)$. That is, if $V \in \mathcal{A}_B$ is a Borel set such that $V \cap \operatorname{Ran}(X) = \emptyset$, then $m_{F_X}(V) = 0$. In this case, we have

$$\int_{\mathbb{R}} g(x) dF_X(x) = \sum_{x \in \operatorname{Ran}(X)} g(x) m_{F_X}(\{x\}) = \sum_{x \in \operatorname{Ran}(X)} g(x) p_X(x). \quad (11)$$

If X is an absolutely continuous random variable, then for any Borel set $V \in \mathcal{A}_B$, we have

$$m_{F_X}(V) = \int_V f_X(x) dx. \quad (12)$$

Therefore, we have for any simple function h ,

$$\int_{\mathbb{R}} h(x) dF_X(x) = \sum_{k=1}^n a_k m_{F_X}(A_k) = \sum_{k=1}^n a_k \int_{A_k} f_X(x) dx \quad (13)$$

$$= \sum_{k=1}^n \int_{A_k} h(x) f_X(x) dx = \int_{\mathbb{R}} h(x) f_X(x) dx. \quad (14)$$

By approximating a general measurable function g by simple functions, we can also show that

$$\int_{\mathbb{R}} g(x) dF_X(x) = \int_{\mathbb{R}} g(x) f_X(x) dx. \quad (15)$$

Therefore, the definition (10) of expectations captures all the cases in (1) and can be applied to any random variable X regardless of its type. In the next section, we will apply this definition to compute the expectation of a singularly continuous random variable. Before that, let us note one useful property of the Lebesgue-Stieltjes integral. For $a, b, c, d \in \mathbb{R}$ with $a < b$, observe that

$$m_{cF+d}((a, b]) = (cF + d)(b) - (cF + d)(a) = c(F(b) - F(a)) = c m_F((a, b]). \quad (16)$$

Therefore, we have the following property of the Lebesgue-Stieltjes integral:

$$\int_{\Omega} g(x) d(cF + d)(x) = c \int_{\Omega} g(x) dF(x). \quad (17)$$

3 Cantor Distribution

Following the construction in [3, Exercise 1.6.48], we define a sequence of functions $\{c_n\}_{n=0}^{\infty}$ on $\mathbb{R}$ as follows:

$$c_0(x) = \begin{cases} 0 & \text{if } x \in (-\infty, 0), \\ x & \text{if } x \in [0, 1], \\ 1 & \text{if } x \in (1, \infty), \end{cases} \quad (18)$$

$$c_n(x) = \begin{cases} 0 & \text{if } x \in (-\infty, 0), \\ \frac{1}{2}c_{n-1}(3x) & \text{if } x \in [0, \frac{1}{3}], \\ \frac{1}{2} & \text{if } x \in (\frac{1}{3}, \frac{2}{3}), \\ \frac{1}{2} + \frac{1}{2}c_{n-1}(3x - 2) & \text{if } x \in [\frac{2}{3}, 1], \\ 1 & \text{if } x \in (1, \infty). \end{cases} \quad (19)$$

See Figure 1 for the graph of the first few elements in this sequence. Observe that c_0 is an increasing and continuous function with $c_0(0) = 0$ and $c_0(1) = 1$. Suppose that c_{n-1} is also an increasing and continuous function with $c_{n-1}(0) = 0$ and $c_{n-1}(1) = 1$ for some $n \geq 1$. Then, observe that $[0, \frac{1}{3}] \ni x \mapsto c_{n-1}(3x) \in [0, 1]$ and $[\frac{2}{3}, 1] \ni x \mapsto c_{n-1}(3x - 2) \in [0, 1]$ are increasing and continuous functions with values 0 and 1 at the endpoints. Therefore, from the definition (19), c_n is also an increasing and continuous function with $c_n(0) = 0$ and $c_n(1) = 1$. By induction, c_n has these properties for all $n \geq 0$. In addition, we claim that

$$|c_n(x) - c_{n-1}(x)| \leq \frac{1}{2^n}, \quad \forall x \in \mathbb{R}, n \geq 1. \quad (20)$$

First, observe that $|c_n(x) - c_{n-1}(x)| = 0$ for all $x \in (-\infty, 0) \cup (\frac{1}{3}, \frac{2}{3}) \cup (1, \infty)$. Next, we have

$$\begin{aligned} |c_1(x) - c_0(x)| &= \begin{cases} \frac{1}{2}x & \text{if } x \in [0, \frac{1}{3}], \\ \frac{1}{2} - \frac{1}{2}x & \text{if } x \in [\frac{2}{3}, 1], \end{cases} \\ &\leq \frac{1}{2}. \end{aligned} \quad (21)$$

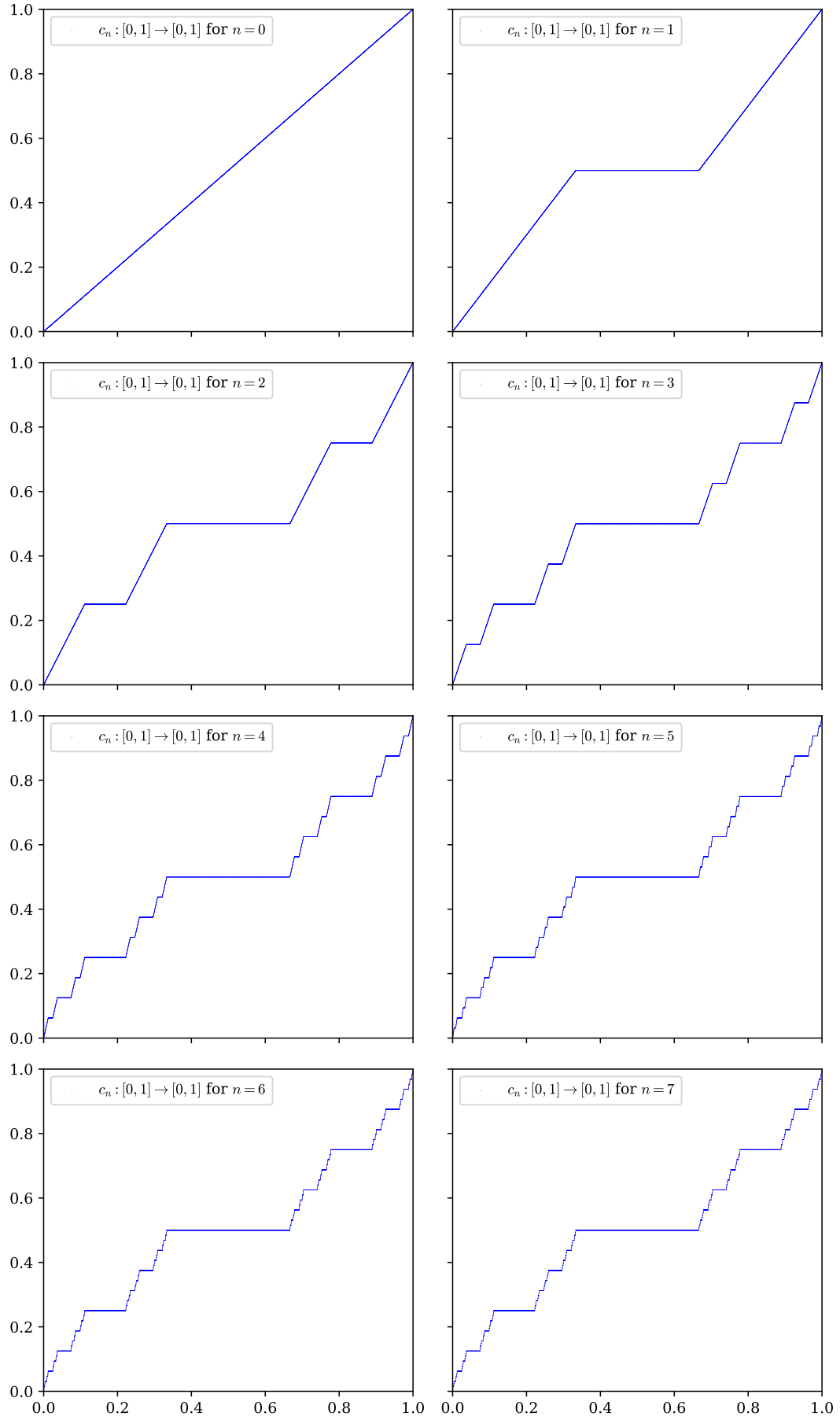


Figure 1: The graphs of c_n for $n = 0, 1, 2, \dots, 7$

Now, suppose that (20) holds for some $n \geq 1$. Then, we have

$$\begin{aligned} |c_{n+1}(x) - c_n(x)| &= \begin{cases} \frac{1}{2} |c_n(3x) - c_{n-1}(3x)| & \text{if } x \in [0, \frac{1}{3}], \\ \frac{1}{2} |c_n(3x-2) - c_{n-1}(3x-2)| & \text{if } x \in [\frac{2}{3}, 1], \end{cases} \\ &\leq \frac{1}{2} \cdot \frac{1}{2^n} = \frac{1}{2^{n+1}}. \end{aligned} \quad (22)$$

Hence, we have proven (20) by induction. As a result, $\{c_n\}$ is a Cauchy sequence in the space $C_b(\mathbb{R})$ of continuous bounded functions on $\mathbb{R}$ equipped with the uniform norm $\|\cdot\|_\infty$ defined by $\|f\|_\infty = \sup_{x \in \mathbb{R}} |f(x)|$ for $f \in C_b(\mathbb{R})$. It is well-known that $C_b(\mathbb{R})$ is complete with respect to $\|\cdot\|_\infty$. Therefore, there exists a continuous bounded function $c : \mathbb{R} \rightarrow \mathbb{R}$ such that $\lim_{n \rightarrow \infty} c_n(x) = c(x)$ for all $x \in \mathbb{R}$. In addition, c is increasing and it satisfies $c(x) = 0$ for $x \in (-\infty, 0]$ and $c(x) = 1$ for $x \in [1, \infty)$. The function c is called the *Devil's staircase* or *Cantor function*.

Using the fact that c is increasing and continuous with $c(x) = 0$ for $x \in (-\infty, 0]$ and $c(x) = 1$ for $x \in [1, \infty)$, we can have a random variable X with the cumulative distribution function $F_X = c$. Although F_X is continuous, it is not absolutely continuous. To show this, we demonstrate that F_X has a zero derivative almost everywhere. First, observe that c_0 has zero derivative outside the set $C_0 = [0, 1]$. Next, c_1 has zero derivative outside the set $C_1 = [0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$. Then, the function c_2 has two copies of c_1 on the intervals $[0, \frac{1}{3}]$ and $[\frac{2}{3}, 1]$. Therefore, c_2 has zero derivative outside the set $C_2 = [0, \frac{1}{9}] \cup [\frac{2}{9}, \frac{1}{3}] \cup [\frac{2}{3}, \frac{7}{9}] \cup [\frac{8}{9}, 1]$. As we continue this process, we obtain C_n by excluding the middle third of each interval in C_{n-1} . Then, the function c has zero derivative outside the set $C = \bigcap_{n=1}^{\infty} C_n$, which is the *Cantor set*. By the construction, the length of each set C_n is

$$\ell(C_n) = \frac{2}{3} \ell(C_{n-1}) = \left(\frac{2}{3}\right)^2 \ell(C_{n-2}) = \dots = \left(\frac{2}{3}\right)^n \ell(C_0) = \left(\frac{2}{3}\right)^n. \quad (23)$$

As a result, the Cantor set C has zero length since

$$\ell(C) \leq \ell(C_n) = \left(\frac{2}{3}\right)^n \xrightarrow{n \rightarrow \infty} 0. \quad (24)$$

Therefore, if F_X can be expressed as an integral of a probability density function f_X , then we must have $f_X = 0$ almost everywhere, which contradicts the fact that F_X is not a zero function. Hence, we have constructed a singularly continuous random variable X .

To compute the expectation $\mathbb{E}(X)$, we cannot use the definition (1) since X is neither discrete nor absolutely continuous. Instead, we can apply the general definition (10) of the expectation as a Lebesgue-Stieltjes integral. Since c is piecewise constant outside the interval $[0, 1]$, which means that $m_{F_X}((-\infty, 0)) = m_{F_X}((1, \infty)) = 0$, we can restrict the integral to $[0, 1]$, i.e.

$$\mathbb{E}(X) = \int_{\mathbb{R}} x dF_X(x) = \int_{[0,1]} x dc(x). \quad (25)$$

To evaluate this integral, we exploit some properties of the Cantor function. From the definition (19), we have for $x \in [0, \frac{1}{3}]$

$$c_n(x) = \frac{1}{2} c_{n-1}(3x) \Rightarrow \lim_{n \rightarrow \infty} \left(c_n(x) - \frac{1}{2} c_{n-1}(3x) \right) = 0 \Rightarrow c(x) = \frac{1}{2} c(3x). \quad (26)$$

Similarly, we have for $x \in [\frac{2}{3}, 1]$,

$$c_n(x) = \frac{1}{2} + \frac{1}{2} c_{n-1}(3x-2) \Rightarrow c(x) = \frac{1}{2} + \frac{1}{2} c(3x-2). \quad (27)$$

Since c is constant on the interval $[\frac{1}{3}, \frac{2}{3}]$, we have

$$\mathbb{E}(X) = \int_{[0, \frac{1}{3}]} x \, dc(x) + \int_{[\frac{2}{3}, 1]} x \, dc(x) \quad (28)$$

$$= \int_{[0, \frac{1}{3}]} x \, d\left(\frac{1}{2}c\right)(3x) + \int_{[\frac{2}{3}, 1]} x \, d\left(\frac{1}{2} + \frac{1}{2}c\right)(3x - 2) \quad (29)$$

$$= \frac{1}{2} \int_{[0, \frac{1}{3}]} x \, dc(3x) + \frac{1}{2} \int_{[\frac{2}{3}, 1]} x \, dc(3x - 2) \quad (30)$$

$$= \frac{1}{2} \int_{[0, 1]} \frac{x}{3} \, dc(x) + \frac{1}{2} \int_{[0, 1]} \frac{x+2}{3} \, dc(x) \quad (31)$$

$$= \frac{1}{3} \int_{[0, 1]} x \, dc(x) + \frac{1}{3} \int_{[0, 1]} dc(x) \quad (32)$$

$$= \frac{1}{3} \mathbb{E}(X) + \frac{1}{3}, \quad (33)$$

where we have done some variable substitutions in (31). As a result, we obtain

$$\mathbb{E}(X) = \frac{1}{2}. \quad (34)$$

Similarly, we can follow the same steps to compute $\mathbb{E}(X^2)$ as follows:

$$\mathbb{E}(X^2) = \int_{[0, 1]} x^2 \, dc(x) \quad (35)$$

$$= \int_{[0, \frac{1}{3}]} x^2 \, dc(x) + \int_{[\frac{2}{3}, 1]} x^2 \, dc(x) \quad (36)$$

$$= \frac{1}{2} \int_{[0, \frac{1}{3}]} x^2 \, dc(3x) + \frac{1}{2} \int_{[\frac{2}{3}, 1]} x^2 \, dc(3x - 2) \quad (37)$$

$$= \frac{1}{2} \int_{[0, 1]} \left(\frac{x}{3}\right)^2 \, dc(x) + \frac{1}{2} \int_{[0, 1]} \left(\frac{x+2}{3}\right)^2 \, dc(x) \quad (38)$$

$$= \frac{1}{9} \int_{[0, 1]} x^2 \, dc(x) + \frac{2}{9} \int_{[0, 1]} x \, dc(x) + \frac{2}{9} \int_{[0, 1]} dc(x) \quad (39)$$

$$= \frac{1}{9} \mathbb{E}(X^2) + \frac{2}{9} \mathbb{E}(X) + \frac{2}{9} \quad (40)$$

$$\implies \mathbb{E}(X^2) = \frac{1}{4} \mathbb{E}(X) + \frac{1}{4} = \frac{3}{8}. \quad (41)$$

Hence, the variance of X is given by

$$\text{Var}(X) = \mathbb{E}(X^2) - \mathbb{E}(X)^2 = \frac{3}{8} - \left(\frac{1}{2}\right)^2 = \frac{1}{8}. \quad (42)$$

In general, we can compute any moment $\mathbb{E}(X^n)$ recursively for $n \geq 1$:

$$\mathbb{E}(X^n) = \int_{[0, 1]} x^n \, dc(x) \quad (43)$$

$$= \int_{[0, \frac{1}{3}]} x^n \, dc(x) + \int_{[\frac{2}{3}, 1]} x^n \, dc(x) \quad (44)$$

$$= \frac{1}{2} \int_{[0, \frac{1}{3}]} x^n \, dc(3x) + \frac{1}{2} \int_{[\frac{2}{3}, 1]} x^n \, dc(3x - 2) \quad (45)$$

$$= \frac{1}{2} \int_{[0, 1]} \left(\frac{x}{3}\right)^n \, dc(x) + \frac{1}{2} \int_{[0, 1]} \left(\frac{x+2}{3}\right)^n \, dc(x) \quad (46)$$

$$= \frac{1}{3^n} \int_{[0, 1]} x^n \, dc(x) + \frac{1}{3^n} \sum_{k=0}^{n-1} \binom{n}{k} 2^{n-k-1} \int_{[0, 1]} x^k \, dc(x) \quad (47)$$

$$= \frac{1}{3^n} \mathbb{E}(X^n) + \frac{1}{3^n} \sum_{k=0}^{n-1} \binom{n}{k} 2^{n-k-1} \mathbb{E}(X^k) \quad (48)$$

$$\implies \mathbb{E}(X^n) = \frac{1}{3^n - 1} \sum_{k=0}^{n-1} \binom{n}{k} 2^{n-k-1} \mathbb{E}(X^k). \quad (49)$$

In the final part of this section, we compute the characteristic function of X defined by

$$\phi_X(t) = \mathbb{E}(e^{itX}) = \int_{[0,1]} e^{itx} \mathrm{d}c(x). \quad (50)$$

By following the same steps as above, we have

$$\phi_X(t) = \int_{[0, \frac{1}{3}]} e^{itx} \mathrm{d}c(x) + \int_{[\frac{2}{3}, 1]} e^{itx} \mathrm{d}c(x) \quad (51)$$

$$= \frac{1}{2} \int_{[0, \frac{1}{3}]} e^{itx} \mathrm{d}c(3x) + \frac{1}{2} \int_{[\frac{2}{3}, 1]} e^{itx} \mathrm{d}c(3x - 2) \quad (52)$$

$$= \frac{1}{2} \int_{[0,1]} e^{it\frac{x}{3}} \mathrm{d}c(x) + \frac{1}{2} \int_{[0,1]} e^{it\frac{x+2}{3}} \mathrm{d}c(x) \quad (53)$$

$$= \frac{1}{2} \phi_X\left(\frac{t}{3}\right) + \frac{1}{2} e^{2it/3} \phi_X\left(\frac{t}{3}\right) \quad (54)$$

$$= e^{it/3} \frac{e^{it/3} + e^{-it/3}}{2} \phi_X\left(\frac{t}{3}\right) \quad (55)$$

$$= e^{it/3} \cos\left(\frac{t}{3}\right) \phi_X\left(\frac{t}{3}\right). \quad (56)$$

By iterating this equation, we have

$$\phi_X(t) = \phi_X\left(\frac{t}{3^n}\right) \exp\left(it \sum_{k=1}^n \frac{1}{3^k}\right) \prod_{k=1}^n \cos\left(\frac{t}{3^k}\right). \quad (57)$$

In the limit $n \rightarrow \infty$, we obtain

$$\phi_X(t) = \phi_X(0) \exp\left(it \sum_{k=1}^{\infty} \frac{1}{3^k}\right) \prod_{k=1}^{\infty} \cos\left(\frac{t}{3^k}\right) = e^{it/2} \prod_{k=1}^{\infty} \cos\left(\frac{t}{3^k}\right), \quad (58)$$

where we used the fact that $\phi_X(0) = 1$ and $\sum_{k=1}^{\infty} \frac{1}{3^k} = \frac{1}{2}$. Note that the infinite product is well-defined even for $t \neq 0$. First, the absolute values of the partial products $\prod_{k=1}^n \cos\left(\frac{t}{3^k}\right)$ are bounded from above by 1. Second, there exists a sufficiently large N such that $|\frac{t}{3^N}| < \frac{\pi}{2}$ and therefore, $\cos\left(\frac{t}{3^n}\right) > 0$ for all $n \geq N$. Hence, the sequence of partial products $\prod_{k=1}^n \cos\left(\frac{t}{3^k}\right)$ for $n \geq N$ is monotonic. As a result, the limit $\prod_{k=1}^{\infty} \cos\left(\frac{t}{3^k}\right)$ exists.

By using the Taylor expansion of $e^{it/2}$ and $\cos\left(\frac{t}{3^k}\right)$, we have

$$\phi_X(t) = \left(1 + \frac{it}{2} - \frac{t^2}{8} + o(t^2)\right) \prod_{k=1}^{\infty} \left(1 - \frac{t^2}{2 \cdot 3^{2k}} + o(t^2)\right) \quad (59)$$

$$= 1 + \frac{it}{2} - \frac{t^2}{2} \left(\frac{1}{4} + \sum_{k=1}^{\infty} \frac{1}{9^k}\right) + o(t^2) \quad (60)$$

$$= 1 + \frac{it}{2} - \frac{t^2}{2} \left(\frac{1}{4} + \frac{\frac{1}{9}}{1 - \frac{1}{9}}\right) + o(t^2) \quad (61)$$

$$= 1 + it\frac{1}{2} - \frac{t^2}{2} \frac{3}{8} + o(t^2). \quad (62)$$

From this expansion, we can read off the first and second moments of X as $\mathbb{E}(X) = \frac{1}{2}$ and $\mathbb{E}(X^2) = \frac{3}{8}$, which are consistent with our previous calculations.

References

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